RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Thursday, 24 September 2026, 10:36 AM
State	Finished
Completed on	Thursday, 24 September 2026, 10:42 AM
Time taken	6 mins 4 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of arr[i] * i, where i is the index of the element (i = 0, 1, 2, ..., N).Write an algorithm based on Greedy technique with a Complexity O(nlogn).

Input Format:

First line specifies the number of elements-n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
#include<stdlib.h>
int compare(const void *a,const void *b)
{
    return(*(int *)a - *(int *)b);
}
int main(){
    int n,i;
    long long sum=0;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
       scanf("%d",&a[i]);
    qsort(a,n,sizeof(int),compare);
    for(i=0;i<n;i++){
        sum+=(long long)a[i] *i;
    }
    printf("%lld",sum);
    return 0;
}
```

	Input	Expected	Got	
✔	5 2 5 3 4 0	40	40	✔

	Input	Expected	Got	
✔	10 2 2 2 4 4 3 3 5 5 5	191	191	✔
✔	2 45 3	45	45	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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